

RAMANUJAN COLLEGE



UNIVERSITY OF DELHI

DATABASE MANAGEMENT SYSTEM Practicals

SEMESTER - 4
(2025-26)

Submitted by	Submitted to
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PRACTICAL 1.

Create and use the following student-society database schema for a college to answer the given (sample) queries using the standalone SQL editor. Here Rollno (ENROLLMENT) and SID (ENROLLMENT) are foreign keys.

```
1  USE StudentSocietyDB;
2  GO
3  CREATE TABLE STUDENT (
4      RollNo CHAR(6) PRIMARY KEY,
5      StudentName VARCHAR(20),
6      Course VARCHAR(10),
7      DOB DATE
8  );
9  CREATE TABLE SOCIETY (
10     SocID CHAR(6) PRIMARY KEY,
11     SocName VARCHAR(20),
12     MentorName VARCHAR(15),
13     TotalSeats INT
14 );
15 CREATE TABLE ENROLLMENT (
16     RollNo CHAR(6),
17     SID CHAR(6),
18     DateOfEnrollment DATE,
19
20     FOREIGN KEY (RollNo)
21     REFERENCES STUDENT(RollNo),
22
23     FOREIGN KEY (SID)
24     REFERENCES SOCIETY(SocID)
25 );
26 INSERT INTO STUDENT VALUES
27 ('R001', 'Aman', 'CS', '2001-05-10'),
28 ('R002', 'Rahul', 'Chem', '2000-03-15'),
29 ('R003', 'Anita', 'CS', '2002-07-20'),
30 ('R004', 'Zoya', 'Math', '2001-01-12'),
31 ('R005', 'Arjun', 'Chem', '2001-09-05'),
32 ('R006', 'Akash', 'CS', '2000-11-18'),
33 ('R007', 'Riya', 'Chem', '2001-02-25');
34
35
36 INSERT INTO SOCIETY VALUES
37 ('S001', 'NSS', 'Gupta', 50),
38 ('S002', 'Debating', 'Sharma', 40),
39 ('S003', 'Dancing', 'Gupta', 30),
40 ('S004', 'Shashakt', 'Verma', 35),
41 ('S005', 'Sports', 'Singh', 45);
42
43
44 INSERT INTO ENROLLMENT VALUES
45 ('R001', 'S001', '2018-01-15'),
46 ('R002', 'S002', '2018-01-20'),
47 ('R003', 'S001', '2018-02-10'),
48 ('R001', 'S003', '2018-02-15'),
49 ('R005', 'S004', '2018-03-01'),
50 ('R006', 'S005', '2018-03-10'),
51 ('R007', 'S002', '2018-03-15'),
52 ('R003', 'S003', '2018-03-20');
```

```

1  USE StudentSocietyDB;
2  GO
3
4  SELECT * FROM STUDENT;
5  SELECT * FROM SOCIETY;
6  SELECT * FROM ENROLLMENT;

```

100 % No issues found

Results Messages

	RollNo	StudentName	Course	DOB
1	R001	Aman	CS	2001-05-10
2	R002	Rahul	Chem	2000-03-15
3	R003	Anita	CS	2002-07-20
4	R004	Zoya	Math	2001-01-12
5	R005	Arjun	Chem	2001-09-05
6	R006	Akash	CS	2000-11-18
7	R007	Riya	Chem	2001-02-25

	SocID	SocName	MentorName	TotalSeats
1	S001	NSS	Gupta	50
2	S002	Debating	Shama	40
3	S003	Dancing	Gupta	30
4	S004	Shashakt	Verma	35
5	S005	Sports	Singh	45

	RollNo	SID	DateOfEnrollment
1	R001	S001	2018-01-15
2	R002	S002	2018-01-20
3	R003	S001	2018-02-10
4	R001	S003	2018-02-15
5	R005	S004	2018-03-01
6	R006	S005	2018-03-10
7	R007	S002	2018-03-15
8	R003	S003	2018-03-20

1. Retrieve names of students enrolled in any society.

```
1  USE StudentSocietyDB;
2  GO
3
4  SELECT DISTINCT StudentName
5  FROM STUDENT S
6  JOIN ENROLLMENT E
7  ON S.RollNo = E.RollNo;
```

100 % No issues found

Results Messages

	StudentName
1	Akash
2	Aman
3	Anita
4	Ajun
5	Rahul
6	Riya

2. Retrieve all society names.

```
1  USE StudentSocietyDB;  
2  GO  
3  
4  SELECT SocName  
5  FROM SOCIETY;
```

100 %

✓ No issues found

Results Messages

	SocName
1	NSS
2	Debating
3	Dancing
4	Shashakt
5	Sports

✓ Query executed successfully.

3. Retrieve students' names starting with the letter 'A'

```
1  USE StudentSocietyDB;  
2  GO  
3  
4  SELECT StudentName  
5  FROM STUDENT  
6  WHERE StudentName LIKE 'A%';
```

100 % No issues found

Results Messages

	StudentName
1	Aman
2	Anita
3	Ajun
4	Akash

Query executed successfully.

4. Retrieve students' details studying in courses 'computer science' or 'chemistry'.

```
1  USE StudentSocietyDB;
2  GO
3
4  SELECT *
5  FROM STUDENT
6  WHERE Course IN ('CS', 'Chem');
```

100 % No issues found

Results Messages

	RollNo	StudentName	Course	DOB
1	R001	Aman	CS	2001-05-10
2	R002	Rahul	Chem	2000-03-15
3	R003	Anita	CS	2002-07-20
4	R005	Arjun	Chem	2001-09-05
5	R006	Akash	CS	2000-11-18
6	R007	Riya	Chem	2001-02-25

Query executed successfully.

5. Retrieve students' names whose roll no either starts with 'X' or 'Z' and ends with '9'

```
1  USE StudentSocietyDB;
2  GO
3
4  SELECT StudentName
5  FROM STUDENT
6  WHERE (RollNo LIKE 'X%' OR RollNo LIKE 'Z%')
7  AND RollNo LIKE '%9';
```

100 % No issues found

Results Messages

StudentName

Query executed successfully.

6. Find society details with more than N TotalSeats where N is to be input by the user

```
1  USE StudentSocietyDB;  
2  GO  
3  
4  SELECT *  
5  FROM SOCIETY  
6  WHERE TotalSeats > 30;
```

100 % No issues found

Results Messages

	SocID	SocName	MentorName	TotalSeats
1	S001	NSS	Gupta	50
2	S002	Debating	Sharma	40
3	S004	Shashakt	Verma	35
4	S005	Sports	Singh	45

Query executed successfully.

7. Update society table for the mentor name of a specific society

```
1  USE StudentSocietyDB;  
2  GO  
3  
4  UPDATE SOCIETY  
5  SET MentorName = 'Singh'  
6  WHERE SocName = 'NSS';
```

100 % No issues found

Messages

(1 row affected)

Completion time: 2026-04-21T23:15:04.2540068+05:30

100 % No issues found

1

2

SELECT * FROM SOCIETY

100 %

No issues found

Results

Messages

	SocID	SocName	MentorName	TotalSeats
1	S001	NSS	Singh	50
2	S002	Debating	Sharma	40
3	S003	Dancing	Gupta	30
4	S004	Shashakt	Vema	35
5	S005	Sports	Singh	45

8. Find society names in which more than five students have enrolled

1	USE StudentSocietyDB;
2	GO
3	
4	SELECT SID
5	FROM ENROLLMENT
6	GROUP BY SID
7	HAVING COUNT(RollNo) > 5;

100 %	No issues found
Results	Messages
SID	

Query executed successfully.

9. Find the name of the youngest student enrolled in society 'NSS'

```
1  USE StudentSocietyDB;  
2  GO  
3  
4  SELECT TOP 1 S.StudentName  
5  FROM STUDENT S  
6  JOIN ENROLLMENT E  
7  ON S.RollNo = E.RollNo  
8  JOIN SOCIETY SO  
9  ON E.SID = SO.SocID  
10 WHERE SO.SocName = 'NSS'  
11 ORDER BY S.DOB DESC;
```

100 % No issues found

Results Messages

	StudentName
1	Anita

Query executed successfully.

10. Find the name of the most popular society (on the basis of enrolled students)

```
1  USE StudentSocietyDB;  
2  GO  
3  
4  SELECT TOP 1 SO.SocName  
5  FROM SOCIETY SO  
6  JOIN ENROLLMENT E  
7  ON SO.SocID = E.SID  
8  GROUP BY SO.SocName  
9  ORDER BY COUNT(*) DESC;
```

100 % No issues found

Results Messages

	SocName
1	Debating

Query executed successfully.

11. Find the name of two least popular societies (on the basis of enrolled students)

```
1  USE StudentSocietyDB;
2  GO
3
4  SELECT TOP 2 SO.SocName
5  FROM SOCIETY SO
6  JOIN ENROLLMENT E
7  ON SO.SocID = E.SID
8  GROUP BY SO.SocName
9  ORDER BY COUNT(*) ASC;
```

100 % No issues found

Results Messages

	SocName
1	Shashakt
2	Sports

Query executed successfully.

12. Find the students names who are not enrolled in any society.

```
1  USE StudentSocietyDB;  
2  GO  
3  
4  SELECT StudentName  
5  FROM STUDENT  
6  WHERE RollNo NOT IN  
7  (  
8      SELECT RollNo  
9      FROM ENROLLMENT  
10 ) ;
```

100 % No issues found

Results Messages

	StudentName
1	Zoya

Query executed successfully.

13. Find the students names enrolled in at least two societies

```
1  USE StudentSocietyDB;  
2  GO  
3  
4  SELECT RollNo  
5  FROM ENROLLMENT  
6  GROUP BY RollNo  
7  HAVING COUNT(SID) >= 2;
```

100 % No issues found

Results Messages

	RollNo
1	R001
2	R003

Query executed successfully.

14. Find society names in which maximum students are enrolled

```
1  USE StudentSocietyDB;  
2  GO  
3  
4  SELECT TOP 1 SID  
5  FROM ENROLLMENT  
6  GROUP BY SID  
7  ORDER BY COUNT(*) DESC;
```

100 % No issues found

Results Messages

	SID
1	S002

Query executed successfully.

15. Find names of all students who have enrolled in any society and society names in which at least one student has enrolled

```
1  USE StudentSocietyDB;
2  GO
3
4  SELECT S.StudentName, SO.SocName
5  FROM STUDENT S
6  JOIN ENROLLMENT E
7  ON S.RollNo = E.RollNo
8  JOIN SOCIETY SO
9  ON E.SID = SO.SocID;
```

100 % No issues found

Results Messages

	StudentName	SocName
1	Aman	NSS
2	Rahul	Debating
3	Anita	NSS
4	Aman	Dancing
5	Ajun	Shashakt
6	Akash	Sports
7	Riya	Debating
8	Anita	Dancing

Query executed successfully.

16. Find names of students who are enrolled in any of the three societies 'Debating', 'Dancing' and 'Sashakt'.

```
1  USE StudentSocietyDB;
2  GO
3
4  SELECT DISTINCT S.StudentName
5  FROM STUDENT S
6  JOIN ENROLLMENT E
7  ON S.RollNo = E.RollNo
8  JOIN SOCIETY SO
9  ON E.SID = SO.SocID
10 WHERE SO.SocName IN ('Debating', 'Dancing', 'Shashakt');
```

100 % No issues found

Results Messages

	StudentName
1	Aman
2	Anita
3	Arjun
4	Rahul
5	Riya

Query executed successfully.

17. Find society names such that its mentor has a name with 'Gupta' in it.

```
1  USE StudentSocietyDB;
2  GO
3
4  SELECT SocName
5  FROM SOCIETY
6  WHERE MentorName LIKE '%Gupta%';
```

100 % No issues found

Results Messages

	SocName
1	Dancing

Query executed successfully.

18. Find the society names in which the number of enrolled students is only 10% of its capacity.

```
1  USE StudentSocietyDB;
2  GO
3
4  SELECT S0.SocName
5  FROM SOCIETY S0
6  JOIN ENROLLMENT E
7  ON S0.SocID = E.SID
8  GROUP BY S0.SocName, S0.TotalSeats
9  HAVING COUNT(E.RollNo) = 0.1 * S0.TotalSeats;
```

100 % No issues found

Results Messages

SocName

Query executed successfully.

19. Display the vacant seats for each society.

```
1  USE StudentSocietyDB;
2  GO
3
4  SELECT
5  SO.SocName,
6  SO.TotalSeats - COUNT(E.RollNo) AS VacantSeats
7  FROM SOCIETY SO
8  LEFT JOIN ENROLLMENT E
9  ON SO.SocID = E.SID
10 GROUP BY SO.SocName, SO.TotalSeats;
```

100 % No issues found

Results Messages

	SocName	VacantSeats
1	Dancing	28
2	Shashakt	34
3	Debating	38
4	Sports	44
5	NSS	48

Query executed successfully.

20. Increment Total Seats of each society by 10%

1
2
3
4
5

```
USE StudentSocietyDB;  
GO  
  
UPDATE SOCIETY  
SET TotalSeats = TotalSeats + (TotalSeats * 0.10);
```

100 %
Messages

No issues found

(5 rows affected)

Completion time: 2026-04-21T23:22:51.7675527+05:30

1
2
3

```
USE StudentSocietyDB;  
GO  
SELECT * FROM SOCIETY;
```

100 %
Results
Messages

	SocID	SocName	MentorName	TotalSeats
1	S001	NSS	Singh	55
2	S002	Debating	Sharma	44
3	S003	Dancing	Gupta	33
4	S004	Shashakt	Verma	38
5	S005	Sports	Singh	49

21. Add the enrollment fees paid ('yes'/'No') field in the enrollment table.

The screenshot displays the SQL Server Enterprise Manager interface. The top pane shows a SQL script with the following commands:

```
1 USE StudentSocietyDB;  
2 GO  
3  
4 ALTER TABLE ENROLLMENT  
5 ADD FeesPaid VARCHAR(3);
```

The bottom pane shows the execution results. A status bar at the top of the results pane indicates "100 %" and "No issues found". Below this, a "Messages" tab is active, displaying the following text:

Commands completed successfully.
Completion time: 2026-04-21T23:35:56.4433945+05:30

At the bottom of the interface, a yellow status bar indicates "Query executed successfully."

22. Update date of enrollment of society id 's1' to '2018-01-15', 's2' to the current date and 's3' to '2018-01-02'.

```
1  USE StudentSocietyDB;
2  GO
3
4  UPDATE ENROLLMENT
5  SET DateOfEnrollment = '2018-01-15'
6  WHERE SID = 'S001';
7
8  UPDATE ENROLLMENT
9  SET DateOfEnrollment = GETDATE()
10 WHERE SID = 'S002';
11
12 UPDATE ENROLLMENT
13 SET DateOfEnrollment = '2018-01-02'
14 WHERE SID = 'S003';
```

100 % No issues found

Messages

(2 rows affected)

(2 rows affected)

(2 rows affected)

Completion time: 2026-04-21T23:36:59.8323648+05:30

100 % No issues found

Query executed successfully.

23. Create a view to keep track of society names with the total number of students enrolled in it.

```
1 USE StudentSocietyDB;
2 GO
3
4 CREATE VIEW SocietyStudentCount AS
5 SELECT
6     SO.SocName,
7     COUNT(E.RollNo) AS TotalStudents
8 FROM SOCIETY SO
9 LEFT JOIN ENROLLMENT E
10 ON SO.SocID = E.SID
11 GROUP BY SO.SocName;
```

100 % No issues found

Messages

Commands completed successfully.

Completion time: 2026-04-21T23:37:29.1801885+05:30


```
1 SELECT * FROM SocietyStudentCount;
```

100 % No issues found

100 % 1 0

Results Messages

	SocName	TotalStudents
1	Dancing	2
2	Debating	2
3	NSS	2
4	Shashakt	1
5	Sports	1

Query executed successfully.

24. Find student names enrolled in all the societies.

```
1  USE StudentSocietyDB;
2  GO
3
4  SELECT StudentName
5  FROM STUDENT S
6  WHERE NOT EXISTS
7  (
8  SELECT SocID
9  FROM SOCIETY
10 WHERE SocID NOT IN
11 (
12 SELECT SID
13 FROM ENROLLMENT
14 WHERE RollNo = S.RollNo
15 )
16 );
```

100 % No issues found

Results Messages

StudentName

25. Count the number of societies with more than 5 students enrolled in it

```
1  USE StudentSocietyDB;
2  GO
3
4  SELECT COUNT(*) AS SocietiesWithMoreThan5Students
5  FROM
6  (
7  SELECT SID
8  FROM ENROLLMENT
9  GROUP BY SID
10  HAVING COUNT(RollNo) > 5
11 ) AS T;
```

100 % No issues found

Results Messages

SocietiesWithMoreThan5Students	
1	0

Query executed successfully.

26. Add column Mobile number in student table with default value '9999999999'

```
1 USE StudentSocietyDB;  
2 GO  
3  
4 ALTER TABLE STUDENT  
5 ADD MobileNo VARCHAR(10)  
6 DEFAULT '9999999999';
```

100 % No issues found

Messages

Commands completed successfully.

Completion time: 2026-04-21T23:42:23.0181572+05:30

100 % No issues found

Query executed successfully.

27. Find the total number of students whose age is > 20 years.

```
1  USE StudentSocietyDB;
2  GO
3
4  SELECT COUNT(*) AS StudentsAge20Plus
5  FROM STUDENT
6  WHERE DATEDIFF(YEAR, DOB, GETDATE()) >= 20;
```

100 % No issues found

Results		Messages
StudentsAge20Plus		
1	7	

Query executed successfully.

28. Find names of students who were born in 2001 and are enrolled in at least one society.

```
1  USE StudentSocietyDB;  
2  GO  
3  
4  SELECT DISTINCT S.StudentName  
5  FROM STUDENT S  
6  JOIN ENROLLMENT E  
7  ON S.RollNo = E.RollNo  
8  WHERE YEAR(S.DOB) = 2001;
```

100 % No issues found

Results Messages

	StudentName
1	Aman
2	Arjun
3	Riya

Query executed successfully.

29. Count all societies whose name starts with 'S' and ends with 't' and at least 5 students are enrolled in the society.

```
1  USE StudentSocietyDB;
2  GO
3
4  SELECT COUNT(*) AS SocietyCount
5  FROM SOCIETY SO
6  JOIN ENROLLMENT E
7  ON SO.SocID = E.SID
8  WHERE SO.SocName LIKE 'S%t'
9  GROUP BY SO.SocID
10 HAVING COUNT(E.RollNo) >= 5;
```

100 % No issues found

Results Messages

SocietyCount

Query executed successfully.

30. Display the following information: Society name ,Mentor name, Total Capacity ,Total Enrolled ,Unfilled Seats

```
1  USE StudentSocietyDB;
2  GO
3
4  SELECT
5  SO.SocName,
6  SO.MentorName,
7  SO.TotalSeats AS TotalCapacity,
8  COUNT(E.RollNo) AS TotalEnrolled,
9  SO.TotalSeats - COUNT(E.RollNo) AS UnfilledSeats
10 FROM SOCIETY SO
11 LEFT JOIN ENROLLMENT E
12 ON SO.SocID = E.SID
13 GROUP BY
14 SO.SocName,
15 SO.MentorName,
16 SO.TotalSeats;
```

100 % No issues found

	SocName	MentorName	TotalCapacity	TotalEnrolled	UnfilledSeats
1	Dancing	Gupta	33	2	31
2	Debating	Sharma	44	2	42
3	NSS	Singh	55	2	53
4	Shashakt	Vema	38	1	37
5	Sports	Singh	49	1	48

Query executed successfully.

II. Do the following database administration commands: Create user, create role, grant privileges to a role, revoke privileges from a role, create index.

```
1 USE StudentSocietyDB;
2 GO
3
4 CREATE LOGIN StudentUser
5 WITH PASSWORD = 'Student@123';
6
7 CREATE USER StudentUser
8 FOR LOGIN StudentUser;
```

100 % No issues found

Messages

Commands completed successfully.

Completion time: 2026-04-21T23:53:39.8341841+05:30

100 % No issues found

Query executed successfully.

```
1 USE StudentSocietyDB;
2 GO
3
4 CREATE ROLE StudentRole;
```

```
1 USE StudentSocietyDB;
2 GO
3
4 GRANT SELECT, INSERT, UPDATE
5 ON STUDENT
6 TO StudentRole;
7
8 GRANT SELECT, INSERT, UPDATE
9 ON SOCIETY
10 TO StudentRole;
11
12 GRANT SELECT, INSERT, UPDATE
13 ON ENROLLMENT
14 TO StudentRole;
```

100 %

✓ No issues found

Messages

Commands completed successfully.

Completion time: 2026-04-21T23:54:40.4353135+05:30

100 %

✓ No issues found

✓ Query executed successfully.

```
1 USE StudentSocietyDB;
2 GO
3
4 EXEC sp_addrolemember
5 'StudentRole',
6 'StudentUser';
```

100 % No issues found

Messages

Commands completed successfully.

Completion time: 2026-04-21T23:55:00.8246965+05:30

100 % No issues found

Query executed successfully.

```
1 USE StudentSocietyDB;
2 GO
3
4 REVOKE UPDATE
5 ON STUDENT
6 FROM StudentRole;
```

100 % No issues found

Messages

Commands completed successfully.

Completion time: 2026-04-21T23:55:00.8246965+05:30

100 % No issues found

Query executed successfully.

```
1 USE StudentSocietyDB;
2 GO
3
4 CREATE INDEX idx_RollNo
5 ON STUDENT(RollNo);
```

100 % No issues found

Messages

Commands completed successfully.

Completion time: 2026-04-21T23:55:35.6535054+05:30

100 % No issues found

Query executed successfully.

```
1 USE StudentSocietyDB;
2 GO
3
4 SELECT name
5 FROM sys.database_principals
6 WHERE type = 'R';
```

100 % No issues found

Results Messages

	name
1	public
2	StudentRole
3	db_owner
4	db_accessadmin
5	db_securityadmin
6	db_ddladmin
7	db_backupoperator
8	db_datareader
9	db_datawriter
10	db_denydatareader
11	db_denydatawriter

Query executed successfully.

```

1  USE StudentSocietyDB;
2  GO
3
4  SELECT
5  dp.name AS RoleName,
6  p.permission_name,
7  p.state_desc
8  FROM sys.database_permissions p
9  JOIN sys.database_principals dp
10 ON p.grantee_principal_id = dp.principal_id
11 WHERE dp.name = 'StudentRole';

```

100 % No issues found

Results Messages

	RoleName	permission_name	state_desc
1	StudentRole	INSERT	GRANT
2	StudentRole	SELECT	GRANT
3	StudentRole	UPDATE	GRANT
4	StudentRole	INSERT	GRANT
5	StudentRole	SELECT	GRANT
6	StudentRole	UPDATE	GRANT
7	StudentRole	INSERT	GRANT
8	StudentRole	SELECT	GRANT
9	StudentRole	UPDATE	GRANT

Query executed successfully.

```

1  USE StudentSocietyDB;
2  GO
3
4  SELECT
5  name AS IndexName,
6  object_name(object_id) AS TableName
7  FROM sys.indexes
8  WHERE name = 'idx_RollNo';

```

100 % No issues found

Results Messages

	IndexName	TableName
1	idx_RollNo	STUDENT

Query executed successfully.

```

1 SELECT * FROM STUDENT;
2 SELECT * FROM SOCIETY;
3 SELECT * FROM ENROLLMENT;

```

100 % No issues found

Results Messages

	RollNo	StudentName	Course	DOB	MobileNo
1	R001	Aman	CS	2001-05-10	NULL
2	R002	Rahul	Chem	2000-03-15	NULL
3	R003	Anita	CS	2002-07-20	NULL
4	R004	Zoya	Math	2001-01-12	NULL
5	R005	Ajun	Chem	2001-09-05	NULL
6	R006	Akash	CS	2000-11-18	NULL
7	R007	Riya	Chem	2001-02-25	NULL

	SocID	SocName	MentorName	TotalSeats
1	S001	NSS	Singh	55
2	S002	Debating	Sharma	44
3	S003	Dancing	Gupta	33
4	S004	Shashakt	Verma	38
5	S005	Sports	Singh	49

	RollNo	SID	DateOfEnrollment	FeesPaid
1	R001	S001	2018-01-15	NULL
2	R002	S002	2026-04-21	NULL
3	R003	S001	2018-01-15	NULL
4	R001	S003	2018-01-02	NULL
5	R005	S004	2018-03-01	NULL
6	R006	S005	2018-03-10	NULL
7	R007	S002	2026-04-21	NULL
8	R003	S003	2018-01-02	NULL

Query executed successfully.